

Eye of Poirot — Methodology v0.2

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Methodology — sanction-exposure metrics (v0.2, frozen 2026-07-13)

This document is the frozen Phase A specification for *Eye of Poirot*. It is written so that an independent analyst can rebuild every published quantity from the event-level tables. Version 0.2 supersedes the draft W1, D1, W2, W2*, DDM, DDMF, and DF nomenclature.

Analysis scope. FIFA World Cups 2014, 2018, and 2022 in full, plus only completed 2026 matches M1–M100. The unplayed 2026 semi-finals, third-place match, and final (M101–M104) are outside this frozen snapshot.

Analysis cohort. Match exposure is reported for every knockout match. Suspension exposure is reported for every team that entered the knockout stage; full-tournament tables may retain eliminated group-stage teams as an auditable secondary cohort.

0. Design goal and claim boundary

A card is converted to **sanction exposure**, measured in minutes. The match metric prices the remaining in-play time affected by a card. The suspension metric also prices the time for which a caution remains capable of triggering a future ban and the nominal time lost when a ban or dismissal is served.

These quantities are descriptive. They do not identify referee intent, causation, or whether an individual foul deserved a card. Score state, foul severity, field position, player behaviour, tactics, and referee assignment remain uncontrolled unless a later registered amendment says otherwise.

1. Data model and sources

1.1 Match, event, and player tables

For edition y , team i , match m , player p , and card event c :

Symbol	Meaning
t_c	cumulative event minute from the source label; 45' +2' becomes 47
$T_{end,m}$	end of the final in-play period, including stoppage and extra time but excluding a penalty shoot-out
F_{im}	fouls committed by team i in match m
F_i	all-tournament fouls committed by team i within the edition cutoff
M_p	nominal minutes played by player p
T_i	sum of nominal match durations for team i : 90 or 120 per match
S_p	nominal minutes unavailable through a dismissal or a suspension being served
I_p	nominal minutes unavailable through a documented injury or illness

The normalized data model consists of match, card, team-match foul, player- match participation, suspension, availability, and evidence tables. Player and team identifiers are retained whenever a source exposes them; normalized names are display fields, not join keys where an identifier exists.

1.2 Source hierarchy

1. FIFA calendar, timeline/event, team-statistics, and match-centre data;
2. FIFA Full-Time Match Reports and tournament technical reports;
3. national federation or club statements;
4. established match reports (including BBC, ESPN, Reuters, and L'Équipe);
5. FotMob, Transfermarkt, or Wikipedia as a documented fallback or cross-check.

FIFA's archived 2014 timelines do not contain usable final-period markers. For 2014 only, $T_{end,m}$ is reconstructed from archived FotMob `matchDetails` period timestamps. In penalty-shoot-out matches, `gameEnded` is not used because it includes the shoot-out; $T_{end,m}$ is 120 plus the announced second-extra-time added time. If that announcement is absent, 120 is retained as an explicit lower bound rather than imputed.

Raw responses and PDFs are held in the private, gitignored archive. Public derived tables carry source URLs, extraction notes, and the snapshot manifest needed to recollect or audit them.

1.3 Event scope

Card type is one of ordinary yellow (Y), second caution in the same match followed by dismissal (Y2), or direct red (R). Cards to coaches or other team officials are retained in the event census but excluded from all player metrics. A card shown during a penalty shoot-out is outside the in-play match metric and is separately flagged.

1.4 Frozen rule and collection references

- [FIFA: 2026 double reset](#)
- [FIFA Council: 2026 regulations amendment](#)
- [FIFA: Russia 2018 disciplinary reminder](#)
- [FIFA: Qatar 2022 suspension explanation](#)
- FIFA calendar and timeline endpoints are recorded per row in the raw manifest and public derived tables.
- FotMob 2014 fallback endpoint pattern: https://www.fotmob.com/api/data/matchDetails?matchId=<match_id>.

2. Match exposure E_m

2.1 Per-card exposure

For a player card shown during an in-play period:

$$x_m(c; \rho) = \rho_c \max(0, T_{end, m(c)} - t_c), \quad \rho_c = \begin{cases} 1, & c = Y, \\ \rho, & c \in \{Y2, R\}. \end{cases}$$

The primary dismissal multiplier is $\rho = 2$. Sensitivity values are $\rho \in \{1, 1.5, 2\}$. A second-yellow event is represented once as Y2; its earlier yellow remains an ordinary yellow event, so no card event is silently discarded.

2.2 Team-match exposure and differentials

$$E_m(i, m; \rho) = \sum_{c: i(c)=i, m(c)=m} x_m(c; \rho).$$

For opponent o :

$$\Delta E_m(i, m) = E_m(i, m) - E_m(o, m).$$

Positive values mean that team i carried more remaining-time card exposure than its opponent. The two team perspectives for a match must be exact opposites.

The foul-normalized quantities are

$$e_m(i, m) = \frac{E_m(i, m)}{F_{im}}, \quad \Delta e_m(i, m) = e_m(i, m) - e_m(o, m).$$

If either foul denominator is zero or missing, the corresponding normalized quantity is missing and is never replaced by zero.

3. Suspension exposure E_s

3.1 Reset windows

A **reset window** is a sequence of matches over which single cautions carry forward.

- 2014, 2018, and 2022: one window from the group stage through the quarter-finals; single cautions are cancelled after the quarter-finals.
- 2026: the group stage is one window and R32–R16–quarter-final is a second window; single cautions are cancelled after the group stage and again after the quarter-finals.

This 2026 treatment follows FIFA's 28 April 2026 rules amendment. A 2026 group card therefore never carries into the knockout stage.

3.2 Nominal in-match remainder

For a caution, define

$$r(c) = \max(0, B_c - t_c), \quad B_c = \begin{cases} 90, & \text{regulation period,} \\ 120, & \text{extra time.} \end{cases}$$

Stoppage-time labels can therefore yield zero nominal remainder. They are not shifted back by one minute.

3.3 Accumulation horizon

Let $N(c)$ be the number of the team’s potential fixtures strictly after the card and before the applicable reset, following the full bracket path. The horizon is not shortened by the team’s actual elimination.

If the player receives a second caution within the same window—either in a later match or as a same-match Y2—the earlier caution’s forward-risk interval stops at that triggering event. A cross-match triggering caution has no forward-risk blocks in that window; it instead creates a served suspension term. Thus a risk interval is never counted beyond the event that realizes it.

Examples before any trigger:

Edition/window	MD1	MD2	MD3	R32	R16	QF	SF/final/third place
2014/2018/2022	4	3	2	—	1	0	0
2026	2	1	0	—	—	—	—
group window							
2026	—	—	—	2	1	0	0
knockout window							

3.4 Player-level sanction minutes

For player p , let C_p^Y be ordinary cautions, C_p^D be dismissal events (Y2 or R), D_p be the number of suspension matches actually served after a dismissal within the observed edition, and A_p be the number of cross-match two-caution suspension matches actually served. Then

$$X_s(p; \rho, \mu) = \sum_{c \in C_p^Y} [r(c) + 90N(c)] + \sum_{c \in C_p^D} \rho \max(0, T_{end, m(c)} - t_c) + 90\mu (D_p + A_p).$$

The primary served-suspension multiplier is $\mu = 1.25$, with sensitivity $\mu \in \{1, 1.25, 1.5\}$. Suspension terms are included only when the loss is observed within the edition cutoff. Pending sanctions after the team’s final observed match are recorded but do not create fictitious served minutes.

3.5 Player opportunity weight ω

$$\omega_p = \frac{M_p}{T_i - |S_p \cup I_p|}.$$

All components use nominal match minutes. **S** and **I** are interval sets and are unioned before subtraction so overlapping reasons cannot be counted twice. The denominator includes only matches in which the player belonged to the tournament squad. Values must satisfy $0 \leq \omega_p \leq 1$; a violation is a data error, not a value to clip silently.

I_p includes a documented in-match injury remainder and documented whole-match absences. A substitution alone is not injury evidence. Every positive injury interval must have a source URL and an evidence note. An unexplained absence stays in the denominator. **S_p** includes the in-match remainder after a dismissal and a suspension actually served.

3.6 Team exposure and normalization

$$E_s(i; \rho, \mu) = \sum_{p \in i} \omega_p X_s(p; \rho, \mu), \quad e_s(i; \rho, \mu) = \frac{E_s(i; \rho, \mu)}{F_i}.$$

The primary denominator is all fouls committed by the team across group and knockout matches within the edition cutoff. A prespecified sensitivity uses knockout-stage fouls only. The edition pooled ratio is

$$\bar{e}_s = \frac{\sum_i E_s(i)}{\sum_i F_i},$$

not the unweighted mean of team ratios.

4. Availability, suspension, and evidence coding

Availability is coded at player $\times$ team-match grain with one of: **played**, **bench**, **suspended**, **injured**, or **unexplained**. Only players with positive unweighted sanction minutes $X_s > 0$ require complete availability coding for the primary build.

1. Participation and bench status come from official line-ups or the best archived match record.
2. Suspensions are derived from the chronological card ledger and checked against the next official line-up.
3. Injury/illness requires explicit evidence. A commentary phrase such as “replaced because of an injury” may establish the in-match remainder; a whole-match absence requires an independent team or media report.
4. If evidence cannot distinguish tactical non-selection from injury, status is **unexplained** and injury minutes are zero.

Each positive injury row contains **player**, **match**, **status**, **interval**, **source_url**, evidence tier, and a short quotation-free paraphrase. The evidence audit is published in `data/sources/injury-evidence.md`.

5. Prespecified sensitivity and quality control

The full grid is

$$\rho \in \{1, 1.5, 2\}, \quad \mu \in \{1, 1.25, 1.5\},$$

crossed with all-tournament versus knockout-only foul denominators. The primary setting is $(\rho, \mu) = (2, 1.25)$ with all-tournament fouls.

Required invariants:

- player-card census: 2014 = 194, 2018 = 224, 2022 = 228, 2026 M1–M100 = 270;
- $E_m \geq 0$, $E_s \geq 0$, and $0 \leq \omega \leq 1$;
- the second-caution stop rule never yields a horizon longer than the reset-only horizon;
- two match perspectives satisfy $\Delta E_m(i, m) = -\Delta E_m(o, m)$, and likewise for Δe_m when defined;
- every positive injury interval has a non-empty source URL;
- 2026 inputs and outputs contain no event from M101–M104.

Hand-checked fixtures cover regulation stoppage, extra time, shoot-outs, direct red, same-match second caution, cross-match accumulation, a reset, and an injury-adjusted opportunity denominator in every edition where applicable.

6. Frozen disclosure and amendments

The analyst had seen earlier exploratory rankings and Argentina-focused draft figures for 2018, 2022, and 2026 before freezing v0.2. Those drafts used the superseded W1/D1/W2/W2* family and included two known rule errors: a semi-final accumulation block after the quarter-final reset, and 2026 group cautions carrying into the knockout stage. They are not confirmatory outputs.

This document, the input manifest, and the analysis cutoff were frozen on 2026-07-13 before 2026 M101. Any later change to a definition, inclusion rule, source hierarchy, parameter, or test must be dated in `docs/AMENDMENTS.md` with its reason and its expected directional effect. Corrections to raw transcription that do not change a rule are logged as data errata. No silent back-editing is permitted after registration.

7. Symbol and output migration

Draft label	v0.2 label / disposition
D1	E_m , exact match exposure
D1 difference	ΔE_m
D1/foul difference	Δe_m
W1	$r(c)$, retained only as a nominal within-match component
W2, W2*, adjusted W2*	superseded by X_s , E_s , and e_s
DDM, DDMF, DF	retired; use sanction_exposure and the symbols above

Canonical machine-readable columns are `exp_match_min`, `exp_susp_min`, `d_exp_match`, `exp_match_per_foul`, `exp_susp_per_foul`, and `omega`. Official figure filenames spell out `match-exposure` or `suspension-exposure`; draft abbreviations do not appear in final figures.

Version history

- **v0.2 (2026-07-13)**. Terminology reset to sanction exposure; unified end-clock convention; formal 2026 two-window reset; second-caution stop rule; primary ρ , μ , denominator, evidence, sensitivity, and amendment rules frozen; scope extended to 2014 and capped at 2026 M100.
- **v0.1 (2026-07-12)**. Exploratory draft. Superseded and retained only for provenance through the archived draft figures.